# Zohaib Ali Mughal

Full-Stack Software Engineer: TypeScript, Next.js, NestJS, Flutter

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## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Dart, SQL, Python
**Frontend:** Next.js, React, Tailwind CSS, TanStack Query, Radix / shadcn UI
**Mobile:** Flutter (BLoC, clean architecture), Capacitor, App Store and Play Store releases
**Backend:** Node.js, NestJS, Fastify, tRPC, REST / OpenAPI, Zod, BullMQ, NATS JetStream
**Data and DevOps:** PostgreSQL, Prisma, Supabase, Redis, Docker, AWS, Railway, Vercel, GitHub Actions
**Payments and Quality:** Stripe Connect, Paymob, Airwallex, Sentry, Jest, Vitest, Playwright, Testcontainers

## PROFESSIONAL EXPERIENCE

**Slidey LLC**, Karachi **Nov 2024 – Present**
Trainee Software Engineer (Nov 2024), Software Engineer I (Mar 2025), Software Engineer II (May 2026)

**ETG Platform** portal.evolutiongroup.net.au | case study **Software Engineer II**

*Field-service platform for an Australian electrical, security and telecom group: admin, client, supplier and technician portals. Built to stop about $5k a week lost to unbilled supplier costs and billing scheduled rather than actual hours.*

- Designed a 25-package Turborepo (Next.js frontend, Fastify API, BullMQ worker): 145 tRPC routers and 1,000+ procedures over a 212-model Prisma schema (dispatch, timesheets, assets, quoting, billing); money as BigInt cents.
- Enforcement-first decision engine: one pure, unit-testable gate checks every dispatch, closeout and invoice transition, returns structured failure codes, and logs every block and manager override (8 live checks on the dispatch calendar).
- Scored two-hop reconciliation of bank transactions to supplier invoices; jobs cannot be invoiced until costs match.
- Supplier integration layer: typed adapters for eight external systems, including three trade portals with no public API (SAP Commerce and Magento storefronts) parsed server-side, plus ABN lookup and two payment feeds.
- Infrastructure: docker-compose local parity, GitHub Actions CI building GHCR images, Railway and Vercel deploys with post-deploy smoke tests; 30,000+ tests across 2,200+ files.

*Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, BullMQ, Redis, Better Auth, Docker.*

**Hyss** hyss.io | case study **Software Engineer I**

*Live-event staffing marketplace: companies hire certified freelance crew, with competency-matched shifts, call sheets, realtime chat, invoicing and payouts. One Next.js codebase serves web, iOS and Android through Capacitor.*

- Type-safe Turborepo: 93 Prisma models, about 340 tRPC procedures, an Android Credential Manager plugin.
- Stripe Connect core: payout onboarding, hybrid seat and usage-metered billing, per-invoice checkout links with three fee-split modes, idempotent webhook reconciliation.
- Two-way Google and Apple (CalDAV) calendar sync with encrypted OAuth tokens and a retryable row-locked job queue; realtime per-event crew chat on Supabase Realtime.

*Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, Supabase, Stripe Connect, Capacitor, Sentry.*

**Conek, for A-SAFE Digital** asafe.com | case study **Software Engineer I**

*Quote-to-cash platform for A-SAFE, integrated with their CRM, drawing and routing tools.*

- Maintained and scaled event-driven microservices on NATS JetStream with type-safe contracts and clean-architecture layering; inter-service latency under 100ms.
- Wrote 500+ unit and integration tests holding 95% coverage, eliminating production regressions.

*Stack: TypeScript, Node.js, Fastify, NATS JetStream, Next.js, React, PostgreSQL, AWS.*

**Sportefy** App Store | Play Store | case study **Trainee Software Engineer**

*Sports booking and matchmaking app live in Pakistan: Flutter on both stores, Next.js admin panel, NestJS API.*

**Flutter app**

- 21 feature modules, 823 passing tests, BLoC with clean architecture; offline request queue with replay, automatic token refresh, fault-tolerant parsing.
- QR check-in, wallet checkout, realtime match chat and notifications on Supabase Realtime; server-driven version gate with hard and soft update prompts.

**Web admin panel and API**

- Next.js admin dashboard over a NestJS API: 48 domain modules, 190+ versioned REST endpoints, additive-only /v1 contract with OpenAPI drift gates in CI; no release broke older installs.
- Payments domain (Paymob behind a pluggable gateway adapter, wallet and credits ledger, promo codes, PDF receipts); multi-actor RBAC from a shared permission matrix; API migrated from EC2 to Railway.

*Stack: NestJS, Flutter, Next.js, Prisma, PostgreSQL, Supabase, Paymob, Railway, Vercel, GitHub Actions.*

## TECHNICAL PROJECTS

**RiskSense, Loan-Default MLOps Pipeline** github **Final-Year Project with Meezan Bank Limited**

- ZenML training pipeline with MLflow experiment tracking predicting loan default on 39,503 distributor-financing records: derived default labels and repayment features, KNN imputation for missing KIBOR tenures, class weighting.

*Stack: Python, ZenML, MLflow, scikit-learn, XGBoost, LightGBM, CatBoost.*

**Automated Crypto-Derivatives Trading System**

- Event-driven TypeScript monorepo (4 services, 8 shared packages): strategy processes publish signals over Redis Streams to one execution service, the only component that holds exchange keys and places orders.
- Every trading limit (sizing, per-coin slots, four exposure ceilings, hedge reserve, margin) in one pure risk function shared by the order gate and the settings screen; a priority-laned rate-limit budget puts orders ahead of market data.

*Stack: TypeScript, Node.js, Fastify, tRPC, Next.js, Prisma, PostgreSQL, Redis Streams, zod, Vitest, Docker.*

**Voice AI Patient-Intake Agent** github

- Phone agent that registers new patients over a U.S. number in English or Spanish: a Retell LLM agent with a 14-section prompt and 5 tool-calling functions, saving to Postgres only after the caller confirms a read-back.

*Stack: TypeScript, Retell AI, OpenAI, Fastify, Zod, Drizzle, PostgreSQL, Next.js, Vitest, Railway, Vercel.*

**Faculty Reviews, Course Review Platform** github **University team project, 2024**

- Course and instructor reviews for IBA students: Flutter app, React web app and a Node.js, Express and MongoDB API (50 endpoints, 6 models); wrote the API, the Flutter app and most of the web app in a three-person team.

*Stack: Flutter, React, Node.js, Express, MongoDB.*

## EDUCATION

**Institute of Business Administration (IBA), Karachi** **Graduated 2025**
BSc in Computer Science. National Talent Hunt Program Scholar (top 33 of 7,290 candidates).

## ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA):** mentored 100+ students through hands-on workshops; class assessment averages rose 30%.
- **Ignite Hackathon:** 1st place for an IoT software prototype built and deployed within a 2-hour limit.
- **Trading Bazaar:** runner-up for algorithmic trading strategies.